# Performance of Aggregation Methods in the HFC RCTA Forecasting Tournament

The goal of this analysis is to evaluate and compare different forecast aggregation methods using data from the HFC RCTA forecasting tournament. We study how various aggregation techniques perform in terms of predictive accuracy and propose an improvement to enhance overall forecasting performance.

## Understanding the Data

Focusing on RCTA, i.e the first year of the competition, we are provided with three main raw data files, namely:-

1. Daily forecasts:-
   This file contains the most recent prediction made by each forecaster before the daily scoring cutoff time. The dataset includes unique identifiers for questions, answers, predictors and other useful information.
   This dataset served as the primary file for the analysis, as it directly aligns with the goal of aggregating forecasts based on the latest prediction for each question-day pair.
2. Questions and Answers:-
   A list of all questions asked throughout the competition along with their answers and associated metadata. The metadata captures each question's descriptive, temporal, and categorical details, including when it was created and resolved, what topic or region it pertains to, and how its possible answers were structured and scored.
3. Prediction sets:-
   This file contains a list of all individual-level forecasts (aka. Prediction sets) created in the system. Basically, it stores all the final forecasts made by each forecaster for each question. It is NOT a per day analysis, but more of a final log of forecasts made by predictors as long as the question was live. Its metadata includes details about when and how forecasts were submitted, the questions and answers they relate to, and contextual information such as rationale, timing, and user identifiers.

## Preprocessing before aggregation

Upon analysis of the daily forecasts file, it becomes evident that a large number of forecasts recorded in the system were simply **system defaults**.
For instance, if a question is multinomial with four possible answers, the **base rate** for each option is ¼ (or 0.25). This means that when only the base rate is recorded for each option, no genuine forecast was actually made. Including these initial default forecasts in our analysis would therefore distort the aggregation results.

Therefore all such default forecasts were dropped, and this reduced the size of our dataset by around 22%.
Apart from this, some generic data-cleaning steps were carried out as safety checks to ensure the dataset was properly structured and error-free. This included parsing and validating date and numeric fields, handling missing entries, converting identifiers to categorical types for efficiency, and rounding forecast values to a fixed decimal precision to prevent false changes caused by floating-point noise.

## Aggregation and Accuracy

The forecasts are now aggregated using the five mentioned methods, along with this, the resolved probability values and ordinal scoring data is imported from the Questions and Answers metadata file.
(*Note: Knowing whether a question is ordinal or not plays an important role in the calculation of Brier Score)*
We observe the following accuracies for each forecasting method

| Method | Brier Score(averaged across each question) |
|---|---|
| Raw Mean | 0.122722 |
| Trimmed Mean | 0.123196 |
| Median | 0.125036 |
| Geometric Mean of Odds | 0.129864 |
| Geometric Mean | 0.136829 |

In Brier Score, *lower* is *better*, meaning raw mean actually outperforms all other methods on average.
From the table we can infer Arithmetic-based methods (Raw Mean, Trimmed Mean, Median) performed better because they balance the crowd's uncertainty and reduce the impact of overconfident or noisy forecasts.

The Raw Mean performs best because it balances the crowd's forecasts evenly, reducing the impact of overconfident or biased individuals. It captures collective knowledge without letting a few strong opinions dominate, resulting in more stable and reliable predictions.
The Trimmed Mean behaves almost the same, since there were few extreme forecasts to remove. It provides a small safeguard against outliers but doesn't change the overall picture much.
The Median is more cautious. By focusing only on the middle forecast, it ignores how strongly forecasters agree, which makes it less responsive and slightly underconfident.

The Geometric Mean and Geometric Mean of Odds, on the other hand, amplify strong or extreme opinions. These methods react more sharply to confident forecasts, which can be useful when those confident predictions are correct, but risky when they're not. In this dataset, the crowd was oftentimes confidently wrong, and geometric methods magnified that overconfidence, leading to poorer accuracy overall.

In short, the arithmetic-based approaches performed better because they maintained balance and resisted the crowd's occasional overconfidence, while the geometric approaches exaggerated it.

## Improved Aggregation Method

Forecaster-Skill Weighted Mean : We weight each forecaster's predictions by their historical accuracy, computed from past resolved questions using the inverse of their average Brier score, and then take a weighted mean of forecasts each day.
(Note: We take the inverse of Brier score as a lower score is better.)

Why it's better: Unlike the baseline methods that treat all forecasters equally, this approach gives more influence to consistently accurate forecasters, improving overall calibration and reducing Brier error.

| Method | Brier Score (averaged across each question) |
|---|---|
| Raw Mean | 0.122722 |
| Trimmed Mean | 0.123196 |
| Median | 0.125036 |
| Geometric Mean of Odds | 0.129864 |
| Geometric Mean | 0.136829 |
| **Forecaster-Skill Weighted Mean** | **0.121900** |

This improvement confirms that incorporating forecaster skill enhances aggregate forecast accuracy.